C.-J. Richard Shi

Associate Professor
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- **Research Interests:** Computer-aided design of VLSI circuits with emphasis on analog/RF system-on-chip design automation, transistor-level circuit analysis and optimization, deep-submicron mixed-signal coupling, mixed-signal testing, and hardware description languages (extended to mixed-technology system modeling and simulation)

- **Teaching Interests:** Undergraduate courses on devices, circuits, software, CAD, circuit simulation, logic, VLSI circuit design; Graduate courses on VLSI, circuits, and CAD.

- **Education**

Institution	Degree Conferred	Date	Study Period
University of Waterloo	Ph.D. in Computer Science	5/94	1/92 – 12/93
University of Waterloo	M.A.Sc. in Electrical Engineering	5/91	9/89 – 12/90
Fudan University	M.Sc. in Electrical Engineering	1/88	9/85 – 12/87
Fudan University	B.S. in Electrical Engineering	7/85	9/81 – 7/85

- **Academic and Industrial Experiences**

Jun. 1998 --- present: Assistant, currently Associate, Professor in Electrical Engineering
University of Washington, Seattle WA
- Built a research program in the area of analog/RF design and test automation with about $5M active external research funding
- On leave in 2000 to start ODT, an analog design automation company with about $1M seed research funding from Air Force and corporate entities

Jun. 1995 --- Jun. 1998: Assistant Professor, University of Iowa, Iowa City, Iowa
- Built a research program in the area of mixed-signal design and test automation with about $1M external research funding.

Jun. 1997 - Aug. 1997: Conexant (Rockwell Semiconductor) Systems, Newport Beach, CA
- Conduced a methodology study with Conexant wireless and mixed-signal divisions identifying design bottleneck and design needs for next 5 years; Introduced AHDL-based design flow: Proposed and prototyped and demonstrated template-driven layout/circuit synthesis.

Jan. 1994 --- Jun. 1995: Senior Engineer, Analogy, Inc., Beaverton OR
- VHDL-AMS definition & VHDL-AMS/Saber simulator architecture

Jan. 1991 --- Dec. 1991: Research Associate, Computer Science, University of Waterloo
- Developed a FSM testing program and applied to memory testing & synthesis of synchronous and asynchronous circuits; formal verification

Jan. 1990 --- Dec. 1990: Design Consultant, VLSI Group, University of Waterloo

Jun. 1998 --- Mar. 1989: Manager, Timing-Analysis, Beijing IC Design Center, China
- Developed an MOS circuit timing analysis & verification tool

Jan. 1998 --- Aug. 1989: Assistant Professor, Fudan University, Shanghai, China
- Hierarchical SPICE (with Kai Zhang, a creator of SPICE2G6), Fast Timing Analysis (Fast Delay Analysis for General Networks).

Technical Awards, Honors and Distinctions

Editorship and Distinctions:

- Associate Editor, IEEE Trans. Computer-Aided Design, Since Dec. 2001
- Associate Editor, IEEE Trans. Circuits and Systems-II, Nov. 1999 to Dec. 2001.
- Member of IEEE 1076.1 VHDL-AMS Language Design Committee
- Founding Chair, IEEE Inter. Workshop on Behavioral Modeling and Simulation

Best Paper Awards:

- IEEE/ACM 36th Design Automation Conference, 1999.
- IEEE 17th VLSI Test Symposium, 1998.
- Shanghai Science and Technology Association, 1987. (1 of 20 papers selected from over 1,000 papers published in 1985-1987 in all scientific research areas.)

Best Paper Award Nominations:

- IEEE International Conference on Computer Design, 2001.
- Asia and South Pacific Design Automation Conference, 1998.
- European Design Automation Conference, 1996.
- Asia and South Pacific Design Automation Conference, 1995

- **Senior Member, IEEE, 1999**.
- **National Science Foundation CAREER Award, 1999.**
- **NSERC Doctoral Prize, 1995.** (Each year two prizes are awarded in engineering by NSERC, Canada's NSF-equivalent research granting agency; the prize consists of a silver medal and a cash award of $5,000.)
- **ACM Distinguished Ph.D. Dissertation Award Nomination, 1995.**
- **Uni. of Waterloo Outstanding Achievement in Graduate Studies Award**, 1994.
- **Science & Technology Promotion Award**, State Education Committee, China, 1988.

Summary of Research Grants (see pages 3-4 for details)

- Principal investigators for over $5.6M active research grants and $1M completed research grants from DARPA, Air Force, NSF, SRC and Industry.

Summary of Publications (see pages 5-10 for details)

- 24 peer reviewed papers; 5 book chapters; 34 fully reviewed conference papers

Summary of Student Supervisions and Teaching (Details available upon request)

- Trained 2 PhDs, 8 Masters, and 19 Undergraduate RAs; Currently supervising 9 PhDs.
- Nominated by students for **Outstanding Educator Award,** College of Engineering, University of Washington, 2000
- Taught circuits and devices, VLSI design, circuit simulation, CAD, logic design courses at both the graduate and undergraduate levels.

References:

- Prof. Ernest Kuh, Department of EECS, University of California, Berkeley (kuh@eecs.berkeley.edu)
- Prof. Anthony Vannelli, Department of ECS, University of Waterloo, Waterloo, Canada (vannelli@panther.uwaterloo.ca)
- Prof. Karti Mayaram, Department of ECE, Oregon State University, OR (karti@ece.orst.edu)
- Prof. Sumit Roy, Department of Electrical Engineering, University of Washington, Seattle WA (roy@ee.washington.edu)

Research Grants –

(C.-J. Richard Shi)

❑ Active Research Grants/Contracts

- **NSF/SRC Joint Initiative on Mixed-Signal Electronic Technologies,** ``*Fast Methods of Coupled Circuit, EM and Logic Simulation"*, Principal Investigator (PI), $450K, (with Prof. Vikram Jandhyala as co-PI), 7/1/2001 --- 6/30/2004

- **Defense Advanced Project Agencies (DARPA)/MTO,** ``*VEGAS --- A Unified Language and Solver for Mixed Signal Multi-Physics Microsystem Simulation"*, Principal Investigator (PI), Phase I $3.6M, (with Livermore and Boeing as subcontractors; UW Portion $2.3M), 7/1/2001 --- 6/30/2004

- **Defense Advanced Project Agencies (DARPA)/MTO,** ``*DDD-based Symbolic Analysis of Analog/RF Circuits"*, PI, $450,000, 7/1/2001- 6/30/2004

- **SRC/Conexant Systems,** *"Automatic Macro Identification for Behavioral Model Insertion for Mixed-Signal System-on-Chip Sign-off Simulation"*, Principal Investigator (PI), $150K (approved and under the contracting process) expected starting date: June 2002.

- **DARPA/MTO, "***CAD Tools for 3-D VLSI Design*", $100K, PI (awarded and under the contracting process) expected starting date: May 2002.

- **National Science Foundation (NSF)/ITR, "***Heterogeneous Systems Integration in System-on-a-Chip Designs*", Co-PI share $328,696, 9/16/2000-9/15/2005 (David Allstot as PI, total $4M for ten co-PIs)

- **NSF/SERG,** ``*Application of Communication-Theoretic Principles to Nano-Interconnect Research"*, PI, $100,000, 9/16/2001-9/15/2003.

- **Semiconductor Research Corporation (SRC),** *"Layout-Dependent Substrate Model Generation for CMOS RF Transistors with Symbolic Model Reduction"*, co-PI share $191,101 (PI: Prof. Karti Mayaram, OSU, total $359,095), 7/1/2000-6/30/2003

- **NSF Career Award,** ``*Behavioral Modeling and Simulation for Mixed-Signal/Mixed-Technology System Design*", PI, $262,000, 3/1/2000 --- 12/31/2004

- **NSF,** *Supplement to "Behavioral Modeling and Simulation for Mixed-Signal/Mixed-Technology System Design": Undergraduate Research Experience Program*, PI, $15,000, 7/1/2000—3/31/2004.

- **University of Washington Royalty Research Fund,** ``*Symbolic Modeling of Deep-Submicron VLSI Interconnect for Performance and Manufacturability*", PI, $26,500, 9/1/2000-2/29/2002.

- **NSF/CDADIC REU Award**, ``*Supplement to Symbolic Modeling and Analysis for Deep-Submicron Analog Design Automation: Undergraduate Research Experience Program", PI*, $5,000, 2001-2002.

- **Air Force Research Laboratory (AFRL)/SBIR,** ``*Genetic Programming and Design Visualization for Analog/RF Design and Modeling"*, PI, $750,000 (awarded to Orora Design Technologies, Inc. Redmond, WA while PI was on leave from UW in 2000)

- **Avant! Corporation, "***Deep-submicron VLSI power/ground network optimization"*, $25,000 (cash gift).

- **Conexant Systems (Formally Rockwell Semiconductor Systems),** ``*Analog/RF Design Automation Research"*, PI, $15,000 (cash gift); ``*AHDL-based Mixed-Signal Synthesis and Test Technology Research"*, PI, $25,000 (cash gift)

Completed Research Grants/Contracts

- **DARPA**, ``*Behavioral Fault Modeling, Test Generation and Design for Test from Mixed-Signal Hardware Description Language VHDL-AMS*", PI, $788,645. 12/20-95-12/19/1999.

- **NSF/CDADIC,** ``*Analog Layout Automation for Performance and Manufacturability—II"*, PI, $60,00, 9/15/1999-9/14/2000

- **NSF/CDADIC**, ``*Analog Layout Automation for Performance and Manufacturability—I*", PI, $52,500, 9/15/1998-9/14/1999.

- **Air Force Research Laboratory/SBIR, "***VHDL-AMS Tools for Mixed-Signal Design"*, $100,000, co-PI (awarded to Orora Design Technologies, Inc.)

- **NSF/CDADIC REU Award,** ``*Supplement to Symbolic Modeling and Analysis for Deep-Submicron Analog Design Automation"*, PI, $5,000, summer 2000.

- **NSF/CDADIC,** ``*Device Generation for Deep-Submicron Analog Design Automation"* PI, $5,000, 6/21/1999-9/21/1999.

- **NASA Space Grant**, ``*AWECAD: A Web-based Environment for Circuit Analysis and Design*, PI, $5,000, 6/21/1999-9/21/1999

- **Air Force Research Laboratory/SBIR,** ``*VHDL-AMS Tools for Mixed-Signal Design"*, $750,000, co-PI (awarded to FTL Systems, Rochester, MN; PI: John Wills)

- **Air Force Research Laboratory/SBIR,** ``*VHDL-AMS Tools for Mixed-Signal Design (Phase I)"*, $100,000, co-PI (awarded to FTL Systems, Rochester, MN; PI: John Wills)

- **NSF,** ``*ATM-based Computing Infrastructure*", $408,695, co-PI (PI: Prof. S.M. Reddy, Co-PIs Profs. T. Casavant, J. Kuhl), 9/1/96 --- 8/30/99.

- **USAF Rome Laboratory,** ``*Analog Hardware Description Languages (VHDL-AMS Standard),* $700,000, Technical Engineer (A contract awarded to MTL Systems, Compass Design Automation and Analogy Inc.), 1994 - 1995

- **Information Technology Research Center of Ontario,** "*Models and Algorithms for Verification and Testing of VLSI Circuits"*, $100,000 (Canadian dollars), co-PI (J.A. Brzozowski as PI, J. Ebergen and H. Jurgensen as co-PIs), 4/1/1991 --- 3/31/1993

Research Publications

(C.-J. Richard Shi)

- **Journal papers published or accepted for publications**

(J-1) X.-D. Tan, C.-J. Shi and F. J.-C. Lee, ``Reliability-constrained area optimization of VLSI power/ground networks via sequence of linear programmings", accepted by *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems.*

(J-2) T. Pi and C.-J Shi, ``Multi-terminal determinant decision diagrams: a new approach to semi-symbolic analysis of large analog circuits", accepted by *IEEE Transactions on Circuits and Systems--II.*

(J-3) C.-J. Shi, ``Finding a minimal test-set for analog diagnostic-dictionary", *IEEE Transactions on Reliability,* accepted with minor revision.

(J-4) C.-J. Shi and W. Tian, ``Efficient fault simulation and test generation of linear analog circuits under parameter variations", *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems,* minor revision under review.

(J-5) W. Tian and C.-J. Shi, ``Nonlinear DC fault simulation: one-step relaxation and adaptive simulation continuation", *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems,* minor revision under review.

(J-6) C.-J. Shi and X.-D. Tan, ``Compact representation and efficient generation of s-expanded symbolic network functions for computer-aided analog circuit design", *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 20, no. 7, pp. 813-827, July 2001.

(J-7) T. Pi and C.-J. Shi, ``Testability analysis of analog circuits via determinant decision diagrams", *IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences,* vol. E83-A, no. 12, Dec. 2000.

(J-8) M. W. Tian and C.-J. Shi, ``Worst case tolerance analysis of linear analog circuits using sensitivity bands", *IEEE Transactions on Circuits and Systems-I: Fundamental Theory and Applications,* vol. 47, no. 8, pp. 1138-1145, August 2000.

(J-9) X.-D. Tan and C.-J. Shi, ``Hierarchical symbolic analysis of large analog integrated circuits via determinant decision diagrams", *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems,* vol. 19, no. 4, pp. 401-412, April 2000.

(J-10) C.-J. Shi and X.-D. Tan, ``Canonical symbolic analysis of large analog circuits with determinant decision diagrams", *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems,* vol. 19, no. 1, pp. 1-18, Jan. 2000.

(J-11) C.-J. Shi and W. Kao, "Guest editorial---special issue on behavioral modeling and simulation of mixed-signal/mixed-technology systems", *IEEE Trans. Circuits and Systems --- II: Analog and Digital Signal Processing,* vol. 46, no. 10, pp. 1261-1261, Oct. 1999.

(J-12) C.-J. Shi and W. Tian, "Simulation and sensitivity of linear analog circuits under parameter variations by robust interval analysis", *ACM Transactions on Design Automation of Electronic Systems,* vol. 4, no. 3, pp. 280-312, July 1999.

(J-13) C.-J. Shi and J. A. Brzozowski, ``A characterization of signed hypergraphs and its applications to VLSI via minimization and logic synthesis", *Discrete Applied Mathematics,* vol. 89, no. 1-3, pp. 223-243, Dec. 1998.

(J-14) C.-J. Shi, ``Entity overloading for mixed-signal abstraction in VHDL", *Journal of Information Science and Engineering*, Special Issue on Hardware Description Languages, vol. 14, no. 3, pp. 633-644, Dec. 1998.

(J-15) N. Godambe and C.-J. Shi, ``Behavioral-level noise modeling and jitter simulation of phase-locked loops with faults using VHDL-AMS", *Journal of Electronic Testing: Theory and Applications (JETTA)*, vol. 9, no. 3, pp. 7-17, August 1998.

(J-16) C.-J. Shi and J. A. Brzozowski, "Cluster-cover: A theoretical framework for a class of VLSI-CAD optimization problems", *ACM Transactions on Design Automation of Electronic Systems*, vol. 3, no.1, pp. 76-107, Jan. 1998.

(J-17) C.-J. Shi, A. Vannelli and J. Vlach, ``Performance-driven layer assignment by integer linear programming and path-constrained hypergraph partitioning", *Journal of Heuristics*, vol. 3, no. 3, pp. 225-243, Nov. 1997.

(J-18) C.-J. Shi and J. A. Brzozowski, "An efficient algorithm for constrained encoding and its applications", *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol.12, no.12, pp.1813-1826, Dec. 1993.

(J-19) C.-J. Shi, "Analysis, sensitivity and macromodeling of the Elmore delay in linear networks for performance-driven VLSI design", *International Journal of Electronics*, vol. 75, no. 3, pp. 467-484, Sept. 1993.

(J-20) C.-J. Shi, A. Vannelli and J. Vlach, ``An improvement on Karmarkar's algorithm for integer programming", *COAL Bulletin of the Mathematical Programming Society*, vol.21, pp. 23-28, 1992.

(J-21) C.-J. Shi, ``A signed hypergraph model of constrained via minimization", *Microelectronics Journal*, vol. 23, no. 7, pp. 533-542, Nov. 1992.

(J-22) J. Vlach, J. Barby, A. Vannelli, I. Talkhan, and C.-J. Shi, "Group delay as an estimate of delay in logic", *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol.10, no.7, pp. 949-953, July 1991.

(J-23) Y. Liu and C.-J. Shi, ``Piecewise delay modeling of MOS VLSI digital circuits", *Research & Progress of Solid State Electronics*, vol. 8, no. 4, pp. 438-442, 1988.

(J-24) C.-J. Shi, ``A method for determining the possible spectral terms of a multi-electron configuration in LS coupling", *Journal of Fudan University: Natural Science Series*, vol. 25, no. 2, pp. 234-238, 1986.

- **Journal papers submitted**

(J-25) V. Jandhyala, Y. Wang, D. Gope, and C.-J. Shi, ``A surface-based integral equation formulation for coupled circuiut-electromagnetic simulation", Submitted to *IEEE Trans. Microwave Theory and Techniques.*

(J-26) X.-D. Tan and C.-J. Shi, ``Automatic generation of interpretable symbolic expressions for small-signal characteristics of analog integrated circuits", submitted to *IEEE Transactions on Circuits and Systems--II.*

(J-27) X.-D. Tan and C.-J. Shi, ``Balanced multi-level multi-way partitioning of analog integrated circuits for hierarchical symbolic analysis", submitted to *ACM Transactions on Design Automation of Electronic Systems.*

(J-28) Alicia Manthe and C.-J. Shi, ``Exact and heuristic minimization of determinant decision diagrams", submitted to *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems.*

(J-29) D. Lungeanu and C.-J. Shi, ``Parallel and distributed simulation of mixed-signal circuits and systems via hybrid and self-adaptive conservative and optimistic scheduling", Working Paper, in preparation for *ACM Transactions on Modeling and Simulation.*

(J-30) Y. Bourai and C.-J. Shi, ``Analog-layout recycling for performance and manufacturability", Working Paper, in preparation for *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems.*

- **Books and peer reviewed book chapters**

  *(Some papers are peer reviewed with the process equivalent to the journal; those marked * later appeared in full version somewhere else)*

(J-31) K. Kundert and C.-J. Shi, *The Designer's Guide to Modeling in Verilog-AMS/VHDL-AMS,* Kluwer Academic Publisher (expected publication date: Dec. 2002).

(J-32) C.-J. Shi, Fault Simulation, Chapter 3 and pp. 55-92 in *Analog and Mixed-Signal Test*, Bapiraju Vinnakota (ed.), Prentice-Hall, 1998.

(J-33) *C.-J. Shi and W. Tian, Simulation and Sensitivity of Linear(ized) Analog Circuit under Parameter Variations, Chapter 44 and pp. 540-551 in VLSI: Integrated Systems on Silicon, Ricardo Reis and Luc Claesen (eds.), Chapman & Hall, 1997.

(J-34) C.-J. Shi and A. Vachoux, VHDL-AMS Design Objectives and Rationale, Chapter 1 and pp. 1-30 in Modeling in Analog Design, vol. 2 in Series Current Issues in Electronic Modeling (CIEM), Jean-Michel Berge, Oz Levia and Jacques Rouillard (eds.), Kluwer Academic Publishers, 1995. (**peer reviewed** with the process equivalent to journal papers)

(J-35) C.-J. Shi, Constrained Via Minimization and Signed Hypergraph Partitioning, pp. 337-356 in Algorithmic Aspects of VLSI Layouts, D. T. Lee and M. Sarrafzadeh (eds.), World Scientific Publishing Company, 1993. (**peer reviewed** with the process equivalent to journal papers)

(J-36) W.-Q. Zhao and C.-J. Shi, Schematic Capture and Automatic Generation of Circuit Diagrams, Chapter 1 and pp. 1-27 in Computer Aided Design of VLSI Circuits: Theory and Methods, P. Tan (ed.), Fudan University Press, 1990.

- **Fully refereed conference proceedings publications**

  (*Most conference papers in this category, especially ICCAD, DAC, VTS, CICC, DATE, EuroDAC, have **typical acceptance ratio 1/3-1/4**; ISCAS, ASP-DAC, and ASIC have acceptance ratio: ½-2/3; those marked by * appear later in journals*)

(C-1) V. Jandhyala, Y. Wang, D. Gope and C.-J. R. Shi, "Coupled electromagnetic-circuit simulation of arbitrarily-shaped conducting structures using triangular meshes," to appear in *Proc. of IEEE 3rd International Symposium on Qualify Electronic Design, March 18-20, 2002, San Jose. CA.*

(C-2) Y. Wang, V. Jandhyala, and C.-J. R. Shi, "Coupled electromagnetic-circuit simulation of arbitrarily-shaped conducting structures," pp. 233-236 in *Proc. of IEEE 10th Topical Meeting on Electrical Performance of Electronic Packaging (EPEP'01),* Oct. 26-31, 2001, Boston, MA.

(C-3) A. Manthe and C.-J. Shi, ``Lower-bound based DDD minimization for efficient symbolic circuit analysis", pp. 374 – 379 in *Proc. IEEE International Conference on Computer Design (ICCD'91),* Austin TX, Oct. 2001. **(Nominated by the program committee for the Best Paper Award)**

(C-4) D. Lungeanu and C.-J. Shi, ``Parallel and distributed VHDL simulation", pp. xxx-xxx in *Proc. IEEE International Conference on Computer Design (ICCD'91),* Austin TX, Oct. 2001.

(C-5) X.-D. Tan and C.-J. Shi, ``Fast power-ground network optimization using equivalent circuit modeling", in *Proc. 38th IEEE/ACM Design Automation Conference (DAC'2001),* Las Vegas, NE, June 2001.

(C-6) *T. Pi and C.-J Shi, ``Multi-terminal determinant decision diagrams: a new approach to semi-symbolic analysis of large analog circuits", pp. 19-22 in *Proc. 37th IEEE/ACM Design Automation Conference (DAC'2000),* Los Angeles, C.A. June 2000. (**154 out of 445 were accepted**)

(C-7) Y. Wang, T. Pi and C.-J. Shi, ``Simple-yet-accurate analytic models for deep-submicron VLSI interconnects", pp. xxx –xxx in *Proc. IFIP International Chip Design Automation Conference,* August 2000.

(C-8) D. Lungeanu and C.-J. Shi, ``Parallel and distributed VHDL simulation", pp.658-662 in *Proc. Design, Automation and Test in Europe (DATE2000),* Paris, France, March 2000. (**106 out of 306 were accepted as regular papers**)

(C-9) Y. Bourai and C.-J. Shi, ``Layout compaction for yield optimization via critical-area minimization", pp. 122-125 in *Proc. Design, Automation and Test in Europe (DATE2000),* Paris, France, March 2000. (**106 out of 306 were accepted as regular papers**)

(C-10) *T. Pi and C.-J. Shi, ``Analog-testability analysis via determinant-decision-diagrams based symbolic analysis", pp. 541-546 in *Proc. Asia and South Pacific Design Automation Conference (ASP-DAC' 00),* Yokohama, Japan, Jan. 2000.

(C-11) X.-D. Tan and C.-J. Shi, ``Symbolic circuit-noise modeling and analysis via determinant decision diagrams", pp. 283-287 in *Proc. Asia and South Pacific Design Automation Conference (ASP-DAC' 00),* Yokohama, Japan, Jan. 2000.

(C-12) *D. Lungeanu and C.-J. Shi, ``Distributed simulation of VLSI systems via lookahead-free self-adaptive optimistic and conservative synchronization", pp. 500-504 in *Proc. IEEE/ACM International Conference on Computer-Aided Design (ICCAD' 99),* San Jose, CA, Nov. 1999. (**106 out of 317 submissions were selected**)

(C-13) *X. Tan, C.-J. Shi, D. Lungeanu, J.-C. Lee, and L.-P. Yuan, ``Area optimization of VLSI power/ground networks via sequence of linear programmings", pp. 78-83 in *Proc. IEEE/ACM 36th Design Automation Conference,* New Orleans, LA, June 1999. (**154 out of 451 were accepted**; **Best Paper Award**)

(C-14) *X. Tan and C.-J. Shi, ``Interpretable symbolic small-signal characterization of large analog circuits using determinant decision diagrams", pp. 448-453 in *Proc. Design, Automation and Test in Europe (DATE' 99),* Munich, Germany, Mar. 10-13, 1999.

(C-15) Y. Bourai and C.-J. Shi, ``Symmetry detection for analog layout recycling", pp. 5-7 in *Proc. Asia and South Pacific Design Automation Conference (ASP-DAC' 99),* Hong Kong, Jan. 18-21, 1999.

(C-16) *X. Tan and C.-J. Shi, ``Balanced multi-level multi-way partitioning of large analog circuits for hierarchical symbolic analysis", pp. 1-4 in *Proc. Asia and South Pacific Design Automation Conference (ASP-DAC' 99),* Hong Kong, Jan. 18-21, 1999.

(C-17) *W. Tian and C.-J. Shi, "Nonlinear DC-fault simulation by one-step relaxation---linear circuit models are sufficient for nonlinear DC-fault simulation", pp. 126-131 in *Proc. IEEE 16th VLSI Test Symposium,* 1998. (**Best Paper Award**)

(C-18) *X. Tan and C.-J. Shi, "Hierarchical symbolic analysis of large analog circuits with determinant decision diagrams", pp. 318-321 in *Proc. IEEE International Symposium on Circuits and Systems*, vol. VI, 1998.

(C-19) W. Tian and C.-J. Shi, "Worst-case analysis of linear analog circuits using sensitivity bands", pp. 110-113 in *Proc. IEEE International Symposium on Circuits and Systems*, vol. VI, 1998.

(C-20) *W. Tian and C.-J. Shi, "Efficient DC fault simulation of nonlinear analog circuits", pp. 899-904 in *Proc. Design, Automation and Test in Europe Conference (DATE)*, 1998 (merged EuroDAC and ED&TC). **(122 out of 448 submissions were accepted)**

(C-21) *C.-J. Shi and W. Tian, "Automated test generation of linear analog circuits under parameter variations", pp. 501-506 in *Proc. Asia and South-Pacific Design Automation Conference*, Tokyo, Japan, Feb. 1998. **(Nominated by the program committee for the Best Paper Award)**

(C-22) *C.-J. Shi and X. Tan, "Symbolic analysis of large analog circuits by determinant decision diagrams", pp. 366-373 in *Proc. IEEE International Conference on Computer-Aided Design (ICCAD'97)*, San Jose, Nov. 1997. **(102 out of 341 were accepted)**

(C-23) *C.-J. Shi and W. Tian, "Non-Monte Carlo simulation and sensitivity of linear(ized) analog circuits under parameter variations", pp. 540-551 in *Proc. IFIP International Conference on Very Large Scale Integration (VLSI'97*), Gramado, BRAZIL, August 26-29, 1997.

(C-24) W. Tian and C.-J. Shi, "Rapid frequency-domain analog fault simulation under parameter tolerances", pp. 275-280 in *Proc. IEEE/ACM Design Automation Conference*, Anaheim CA, June 9-13, 1997. (**139 out of 389 accepted**)

(C-25) *N. Godambe and C.-J. Shi, "Behavioral-level noise modeling and jitter simulation of phase-locked loops with faults using VHDL-AMS", pp. 177-183 in *Proc. IEEE VLSI Test Symposium (VTS'97)*, Monterey CA, April 27-30, 1997. **(62 out of 171 accepted)**

(C-26) C.-J. Shi, "Block-level fault isolation using partition theory and logic minimization techniques", pp.319-324 in *Proc. Asia and South Pacific Design Automation Conf. (ASP-DAC'97*), Chiba, Japan, Jan. 1997.

(C-27) C.-J. Shi, "Solving constrained via minimization by compact linear programming", pp.635-640 in *Proc. Asia and South Pacific Design Automation Conf.*, Chiba, Japan, Jan. 1997.

(C-28) O. Coudert and C.-J. Shi, "Exact dichotomy-based constrained encoding", pp.426-431 in *Proc. IEEE International Conf. on Computer Design (ICCD'96)*, Austin, TX, Oct. 1996.

(C-29) C.-J. Shi and N. Godambe, "Behavioral fault modeling and simulation of phase-locked loops using a VHDL-AMS like language", pp.171-174 in *Proc. IEEE International ASIC Conf. & Exhibit (ASIC'96)*, Rochester, NY, Sept. 23-27, 1996. (acceptance ratio: 0.5)

(C-30) *C.-J. Shi, A. Vannelli and J. Vlach, "Performance-driven layer assignment for printed circuit boards and integrated circuits", pp.245-250 in *Proc. IEEE International ASIC Conf. & Exhibit (ASIC'96)*, Rochester, NY, Sept. 23-27, 1996. (acceptance ratio: 0.5)

(C-31) *C.-J. Shi, "Entity overloading for mixed-signal abstraction in VHDL", pp. 562-567 in *Proc. European Design Automation Conf. (EuroDAC/EuroVHDL'96)*, Sept. 16-20, 1996, Geneva, Switzerland**. (Nominated by the program committee for the Best Paper Award)**

(C-32) O. Coudert and C.-J. Shi, "Exact multi-layer topological planar routing", pp. 179-182 in *Proc. IEEE Custom Integrated Circuit Conf. (CICC'96*), San Diego, CA, May 5-8, 1996.

(C-33) *C.-J. Shi and J. A. Brzozowski, "A framework for the analysis and design of algorithms for a class of VLSI optimization problems", pp. 67- 74 in *Proc. Asia and South Pacific Design Automation Conf. (ASP-DAC'95*), Aug. 29 - Sept. 1, 1995. **(Nominated by the program committee for the Best Paper Award)**

(C-34) C.-J. Shi, E. Christen, P. Liebmann, S. Krolikoski, and W. Zhou, "VHDL-AMS: Analog extension to VHDL", pp. 160-165 in *Proc. IEEE International ASIC Conf. & Exhibit*, Rochester, N.Y., Sept. 19-23, 1994.

(C-35) *C.-J. Shi and J. A. Brzozowski, "Efficient constrained encoding for VLSI sequential logic synthesis", pp. 266- 271 in *Proc. First European Design Automation Conf.*, Hamburg, Germany, Sept. 1992. **(79 papers out of 335 were accepted)**

(C-36) *C.-J. Shi, "A signed hypergraph model of constrained via minimization", pp. 159-166 in *Proc. Second Great Lakes Symp. on VLSI*, Kalamazoo, MI, Feb. 1992.

(C-37) C.-J. Shi, "Modeling k-way splits of physical routings for constrained via minimization", pp.4B.5.1-4B.5.8 in *Proc. Canadian Conf. on VLSI*, Aug. 1991.

(C-38) *C.-J. Shi, A. Vannelli, and J. Vlach, "A hypergraph partitioning approach to the via minimization problem", pp. 2.7.1-2.7.8 in *Proc. Canadian Conf. on VLSI*, Oct. 1990.

(C-39) C.-J. Shi and K. Zhang, "Tree relaxation: a new iterative solution method for linear equations", pp. 2355-2359 in *Proc. IEEE Int. Symp. on Circuits and Systems (ISCAS'88*), June 1988.

(C-40) *C.-J. Shi and K. Zhang, "A robust approach to timing verification", pp. 56-59 in *Proc. IEEE Int. Conf. Computer-Aided Design (ICCAD'87)*, Santa Clara, CA, Nov. 1987 **(119 papers out of 440 were accepted: this paper has been widely cited)**

(C-41) C.-J. Shi and K. Zhang, "A newly defined signal delay time and its basic theory", pp. 142-146 in *Proc. IEEE Asian Electronics Conf*., Hong Kong, Sept. 1987.

- **Non-journal abstract refereed publications**

*(We include in this category those fully refereed papers published in workshops, papers in those conferences that accept most papers submitted, and papers published in Chinese when the author was in China where some conferences actually had a very high reject rate; extended versions of those marked by * later appear in journals)*

(C-42) C.-J. Shi, Y. Ye and X. Tan, "Behavioral model optimization via sensitivity-enhanced genetic search", pp. 17-24 in Proc. *IEEE/ACM International Workshop on Behavioral Modeling and Simulation*, Washington DC, Oct. 1997.

(C-43) C.-J. Shi, "Fault isolation of mixed-signal systems using blankets", *IEEE/IMAPS MCM Test IV Workshop*, Napa Valley, California, Sept. 14 - 17, 1997.

(C-44) C.-J. Shi, "Fault isolation for mixed-signal multichip modules", *IEEE MCM Test II Workshop*, Napa Valley, California, Sept. 15 - 18, 1996.

(C-45) C.-J. Shi, "Finding a minimal test set for analog fault diagnostic dictionary", pp. 303-308 in *Proc. International Workshop on Computer-Aided Design, Test, and Evaluation for Dependability*, Beijing, China, July 1996.

(C-46) C.-J. Shi, "Inclusive-OR Boolean satisfiability for via minimization and constrained encoding in VLSI design", *DIMACS Workshop on Satisfiability Problem: Theory and Applications*, March 11-13, 1996.

(C-47) C.-J. Shi, "Towards a unified operational semantics for behavior specification of VLSI systems", *Proc. International Workshop on Computer-Aided Co-Design*, Cambridge, MA, Oct. 7-8, 1993.

(C-48) *C.-J. Shi and A. Vannelli, "An improvement on Karmarkar's algorithm for integer programming", in *Proc. Sixth SIAM Conf. on Discrete Mathematics*, Vancouver, June 1992.

(C-49) C.-J. Shi and K. Zhang, ``A new approach to delay analysis in RC networks", *Proc. 1st National Conference on IC-CAD,* Harbin, China, April 1997.

(C-50) C.-J. Shi, Y. Liu and J. Fang, ``Design of a fast multiplexer using Fudan VLSI-CAD tools", Proc. Chinese National Conference on ICCAD, Guiliangm China, Aug. 1986.

(C-51) C.-J. Shi, ``New criterion of latchup effects in CMOS circuits", *Proc. Chinese National Conference on ICCAD*, Guiliang, China, Aug. 1986.

(C-52) C-.J. Shi and K. Zhang, ``HTA: A hierarchical tearing approach and its implementation in SPICE 2G6", *Proc. 4th National Design Automation Conference*, Shenzen, China, Nov. 1985.

(C-53) C.-J. Shi, L.-Y. Yu and K. Zhang, ``CAP: A general-purpose circuit analysis program on IBM-PC", *Proc. Chinese Symposium on Circuits and Systems*, Changsha, China, Oct. 1985. (**Best Paper Award)**

Professional Services –

(C.-J. Richard Shi)

- **Editorship, Professional society and technical committee services**
  - Associate Editor, *IEEE Transactions on Computer-aided Design of Integrated Circuits and System*, since Dec. 2001.
  - Associate Editor, *IEEE Transactions on Circuits and Systems—II* Analog and Digital Signal Processing, Nov. 1999 to Dec. 2001.
  - Guest Editor, *IEEE Transactions on Circuits and Systems—II* Analog and Digital Signal Processing, special issue on Behavioral Modeling and Simulation of Mixed-Signal/Mixed-Technology Systems, Oct. 1999.
  - Members of Technical Program Committees for
    - IEEE International Conference on Computer Design (ICCD), 2000, 2001
    - IEEE International Symposium on Circuits and Systems (ISCAS), 2000.
    - IEEE International Mixed-Signal Testing Workshop, 1997, 1998, 1999, 2000, 2001
    - IFIP International Conference on Chip Design Automation (ICDA), 2000.
    - Fall VIUF Conference, 1998, Spring VIUF, 1998; International Hardware Description Language Conference, 1999.
  - Founder and served as the Program Chair for IEEE/ACM International Workshop on Behavioral Modeling and Simulation, 1997, 1998 and 1999.
  - IEEE Design Automation Standards (DASC) services (1994-present): was one of the original contributor to, and promoters of, VHDL-AMS; which was approved by IEEE in March 1999 as IEEE std 1076.1-1999
    - Served on IEEE DASC 1076,1 Ballot Review Committee
    - Served on IEEE DASC 1076.1 Language Design Committee
    - Served on IEEE DASC 1076.1 Language Requirement Analysis Committee
    - Was the Principal Author of *VHDL-AMS Design Objectives and Rational Document*, approved by IEEE DASC 1076.1 Working Group as the official document for 1076.1 standardization effort

- **Conference Tutorial and Panel Speakers**

  - Panelist, ``Are We Ready for Analog Circuit Synthesis?", IEEE/ACM/VIUF International Workshop on Behavioral Modeling and Simulation, Oct. 2000
  - Panelist, ``Taming the Beast of Analog Fault Simulation", IEEE International Mixed-Signal Testing Workshop, May 1999.
  - Tutorial Speaker, International VHDL/Verilog Conference, March 1998.
  - Tutorial Speaker, Asia and South Pacific Design Automation Conference, Tokyo, Japan, Feb. 1998.
  - Tutorial Speaker, IEEE International Mixed-Signal Testing Workshop, Seattle, Washington, June 1997
  - Tutorial Speaker, European Design Automation Conference, Geneva, September 19, 1996.
  - Tutorial Speaker, IEEE/ACM 32nd Design Automation Conference, San Francisco, CA, June 12-16, 1995.
  - Tutorial Speaker, IEEE International ASIC Conference and Exhibit, Austin, TX, September 23, 1995.

- Tutorial Speaker, International Conference on Electronic Hardware Description Languages, Las Vegas, Jan. 1995.
- Tutorial Speaker, IEEE International ASIC Conference and Exhibit, Rochester NY, Oct. 1994.
- Tutorial Speaker, IEEE Custom Integrated Circuits Conference, San Diego, May 1994.

- **Others (referee for funding agencies, journals and conferences)**

Invited Participant, DARPA Workshop on Next Generation CAD Tools for Integrated Mixed Technology Systems-on-a-Chip, Oct. 5, 2000.

Panelist, NSF CISE Design Automation CAREER Award Panel, October 2000

Panelist, NSF CISE Design Automation Program Panel, April 2000

Final Reviewer, IEEE/ACM DAC Best Paper Award Selection, March 2000

External Ph.D. Examiner, University of Waterloo, Canada, Oct. 1999

Reviewer for the following funding agencies

- Engineering and Physical Sciences Research Council, Europe (1 proposal), 1999
- National Science Foundation (NSF): many proposals (1999-present)
- National Science and Engineering Research Council (NSERC), Canada
- University of California MICRO Program (3 Proposals), June 2000, June 2001
- University of Washington RRF Program

Reviewer for the following journals and publishers:

- IEEE Transactions for Computer-Aided Design of Integrated Circuits and Systems (total of 25 papers)
- IEEE Transactions on Circuits and Systems (total of 20 papers)
- ACM Transactions on Design Automation of Electronic Systems (1 paper)
- Discrete Applied Mathematics (3 papers)
- International Journal of Analog Integrated Circuits and Signal Processing
- Kluwer Academic Publishers (1 book)
- Irwin McGraw-Hill (1 book)
- World Scientific Publisher (1 book chapter)
- IEEE/ACM Design Automation Conference
- IEEE VLSI Test Symposium
- International Test Conference
- IEEE International Workshop on Behavioral Modeling and Simulation
- International Verilog Conference/VHDL International User Forum
- International Hardware Description Languages
- FDL'99
- IFIP Conference on Computer Hardware Description Languages
- European Conference on Circuit Theory and Design
- Canadian Conference on Very Large Scale Integration
- Workshop on New Directions in Testing
- IEEE Mixed-Signal Testing Workshop